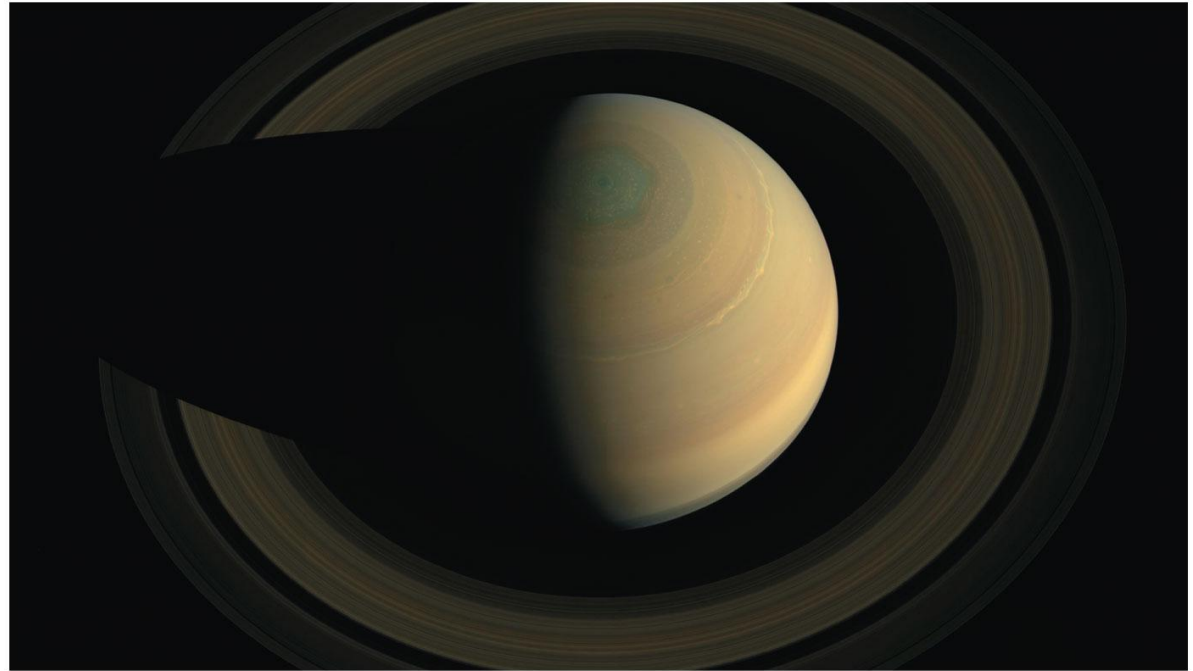


Chapter 6

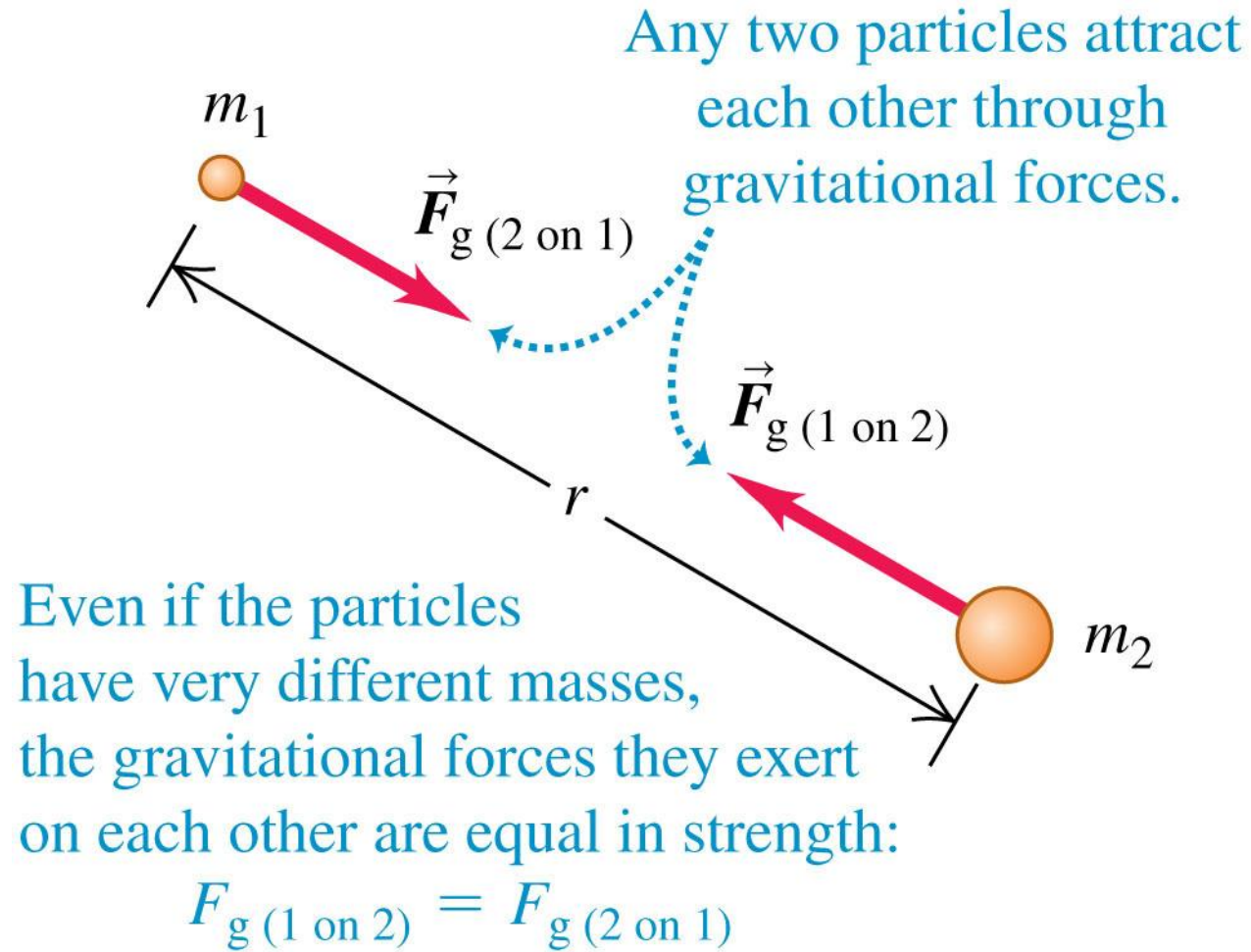
Gravity

Introduction

- What can we say about the motion of the particles that make up Saturn's rings?
- Why doesn't the moon fall to earth, or the earth into the sun?
- By studying gravitation and celestial mechanics, we will be able to answer these and other questions.



Newton's law of gravitation



Newton's law of gravitation

- *Law of gravitation:* Every particle of matter attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them:

Newton's **law of gravitation:**
Magnitude of attractive gravitational force between any two particles

Gravitational constant (same for any two particles)

Masses of particles

Distance between particles

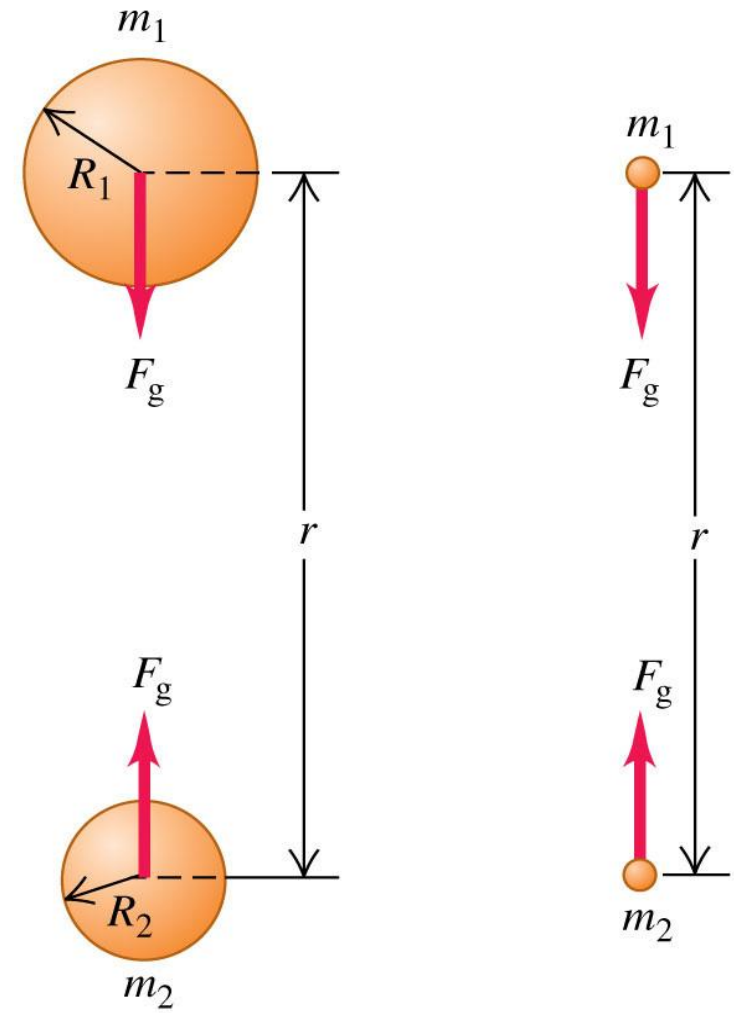
$$F_g = \frac{Gm_1m_2}{r^2}$$

The diagram shows the equation $F_g = \frac{Gm_1m_2}{r^2}$ with four labels and arrows pointing to the corresponding parts: 'Magnitude of attractive gravitational force between any two particles' points to F_g ; 'Gravitational constant (same for any two particles)' points to G ; 'Masses of particles' points to m_1m_2 ; and 'Distance between particles' points to r^2 .

- The gravitational constant G is a fundamental physical constant that has the same value for any two particles.
- $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$.

Gravitation and spherically symmetric bodies

- The gravitational effect *outside* any spherically symmetric mass distribution is the same as though all of the mass were concentrated at its center.
- The force F_g attracting m_1 and m_2 on the left is equal to the force F_g attracting the two point particles m_1 and m_2 on the right, which have the same masses and whose centers are separated by the same distance.



Gravitational attraction

- Our solar system is part of a spiral galaxy like this one, which contains roughly 10^{11} stars as well as gas, dust, and dark matter.
- The entire assemblage is held together by the mutual gravitational attraction of all the matter in the galaxy.



Weight

- The **weight** of a body is the total gravitational force exerted on it by all other bodies in the universe.
- At the surface of the earth, we can neglect all other gravitational forces, so a body's weight is:

Weight of a body at the earth's surface ...
... equals gravitational force the earth exerts on body.

$$w = F_g = \frac{Gm_E m}{R_E^2}$$

Gravitational constant G
Mass of the earth m_E
Mass of body m
Radius of the earth R_E

- The acceleration due to gravity at the earth's surface is:

Acceleration due to gravity at the earth's surface

$$g = \frac{Gm_E}{R_E^2}$$

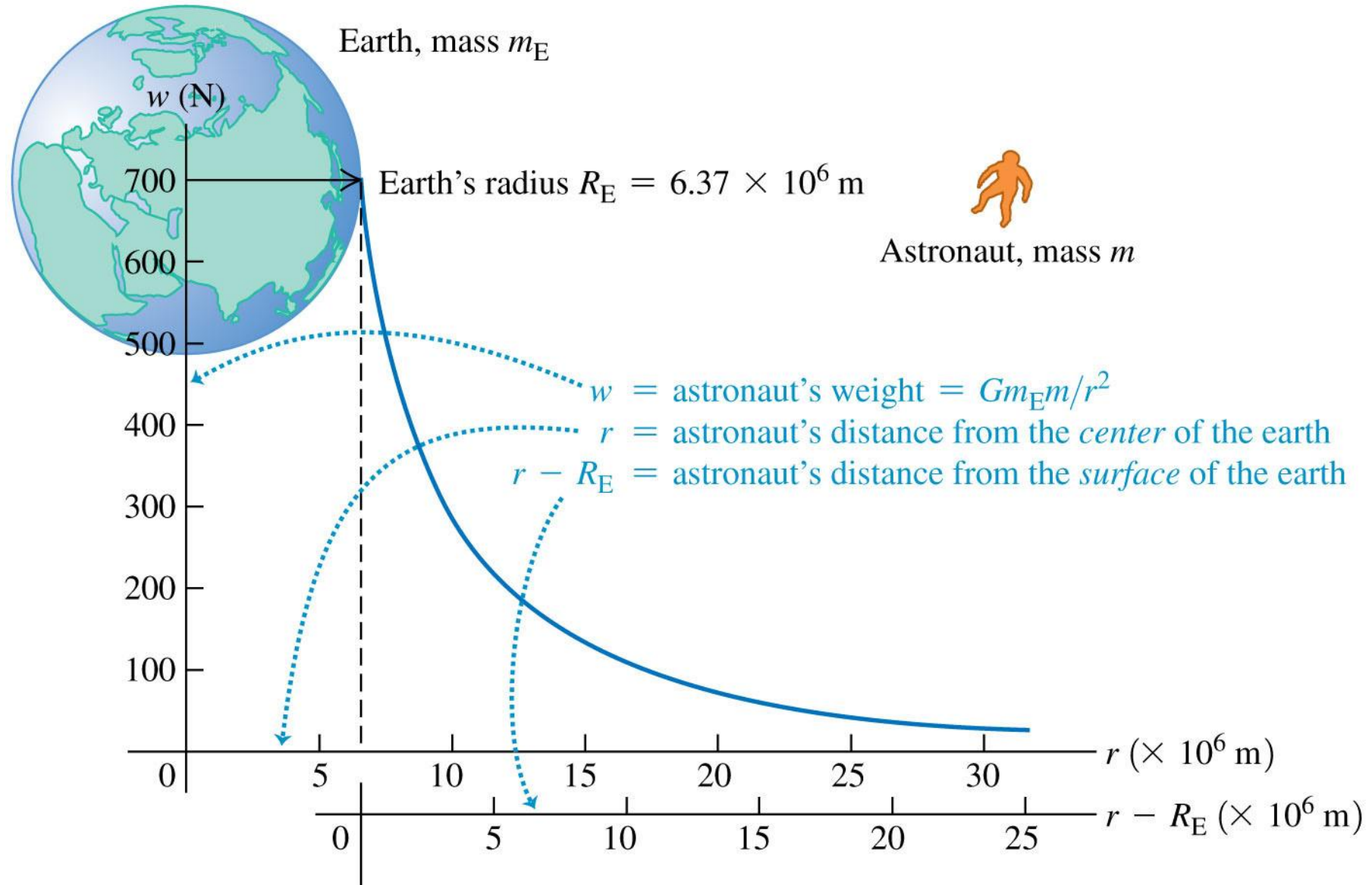
Gravitational constant G
Mass of the earth m_E
Radius of the earth R_E

Walking and running on the moon



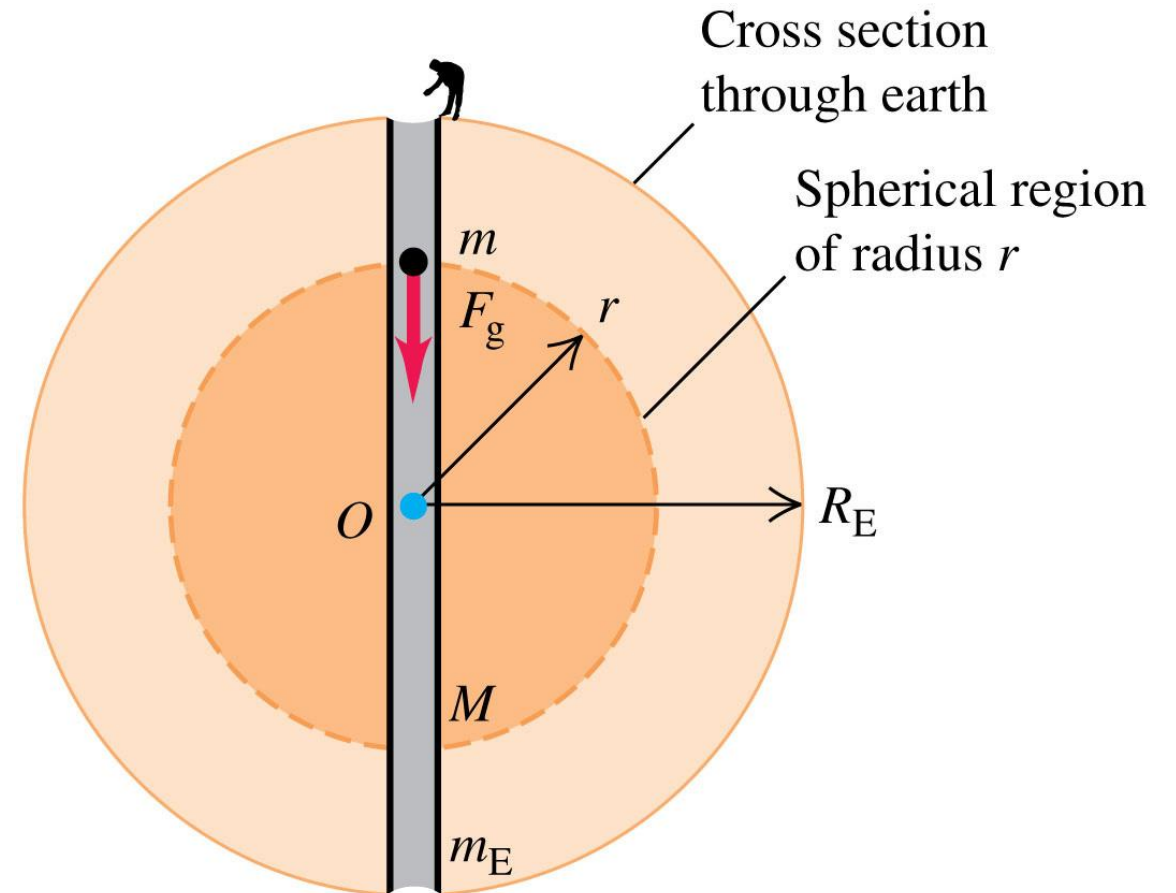
- You automatically transition from a walk to a run when the vertical force the ground exerts on you exceeds your weight.
- This transition from walking to running happens at much lower speeds on the moon, where objects weigh only 17% as much as on earth.
- Hence, the Apollo astronauts found themselves running even when moving relatively slowly during their moon “walks.”

Weight decreases with altitude

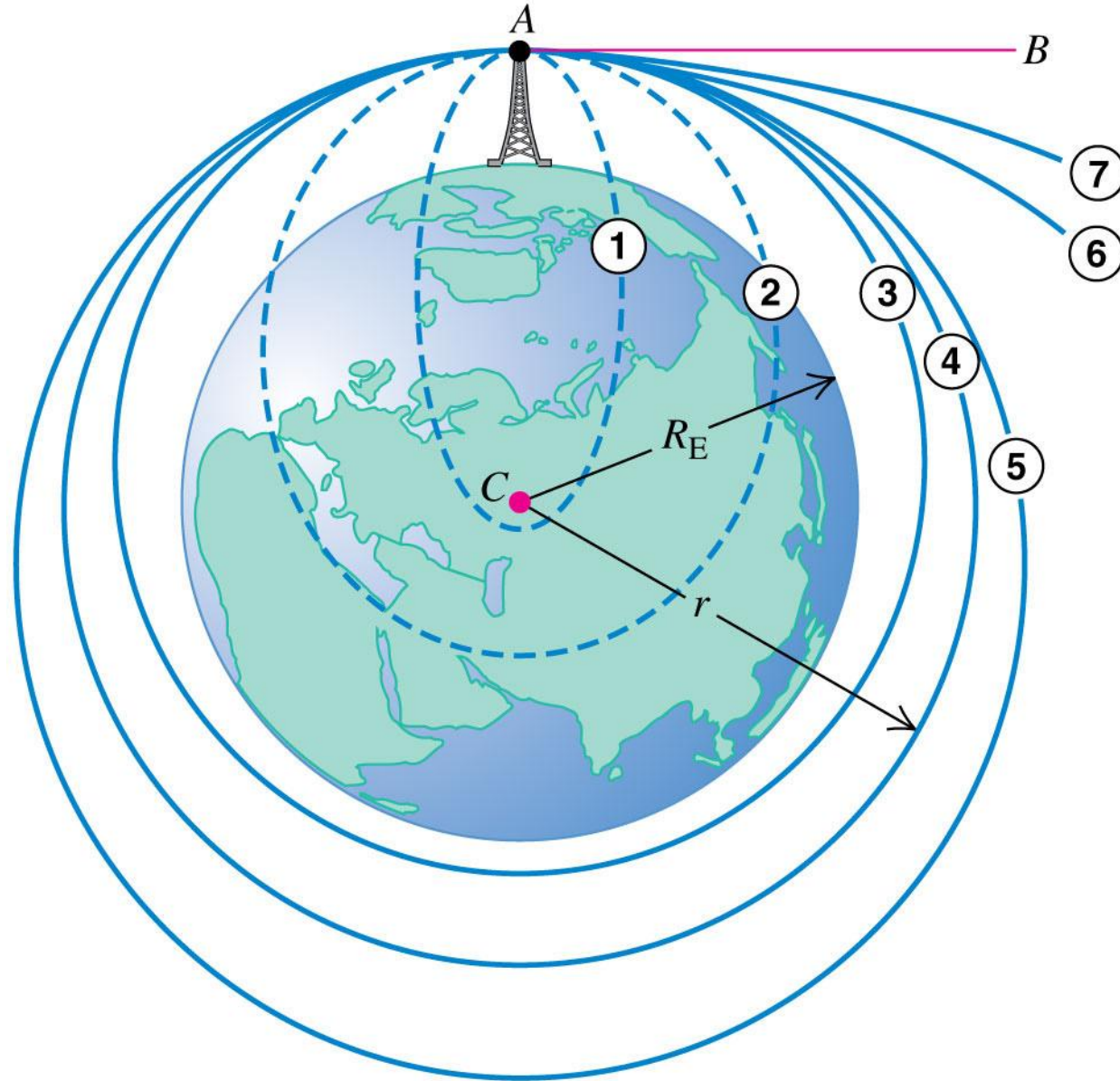


A point mass inside a spherical shell

- If a point mass is inside a spherically symmetric shell, with radius r from the center O .
- Only the mass inside a sphere of radius r exerts a net gravitational force on it.



The motion of satellites



A projectile is launched from A toward B. Trajectories ① through ⑦ show the effect of increasing initial speed.

- The trajectory of a projectile fired from a great height (ignoring air resistance) depends on its initial speed.

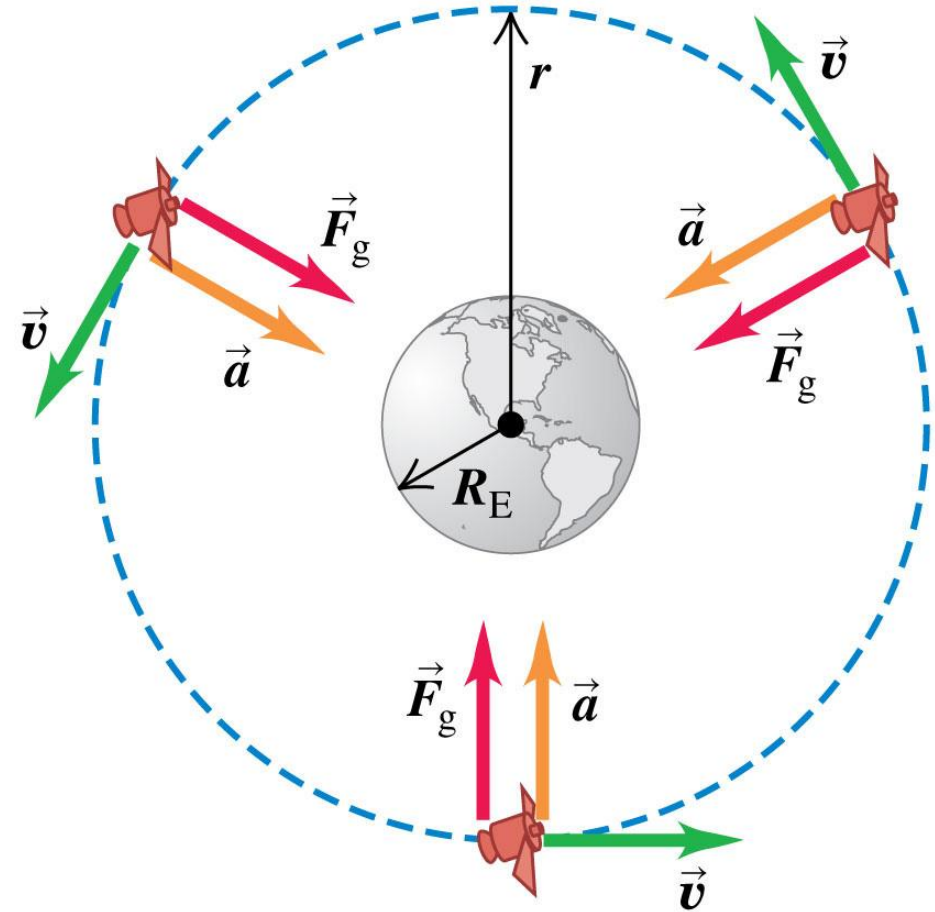
Circular satellite orbits

- With a mass of approximately 4.5×10^5 kg and a width of over 108 m, the International Space Station is the largest satellite ever placed in orbit.



Circular satellite orbits

- For a circular orbit, the speed of a satellite is just right to keep its distance from the center of the earth constant.
- The force $\vec{F}_g$ due to the earth's gravitational attraction provides the centripetal acceleration that keeps a satellite in orbit.



The satellite is in a circular orbit: Its acceleration $\vec{a}$ is always perpendicular to its velocity $\vec{v}$, so its speed v is constant.

Circular satellite orbits

- A satellite is constantly falling *around* the earth.
- Astronauts inside the satellite in orbit are in a state of *apparent weightlessness* because they are falling with the satellite.

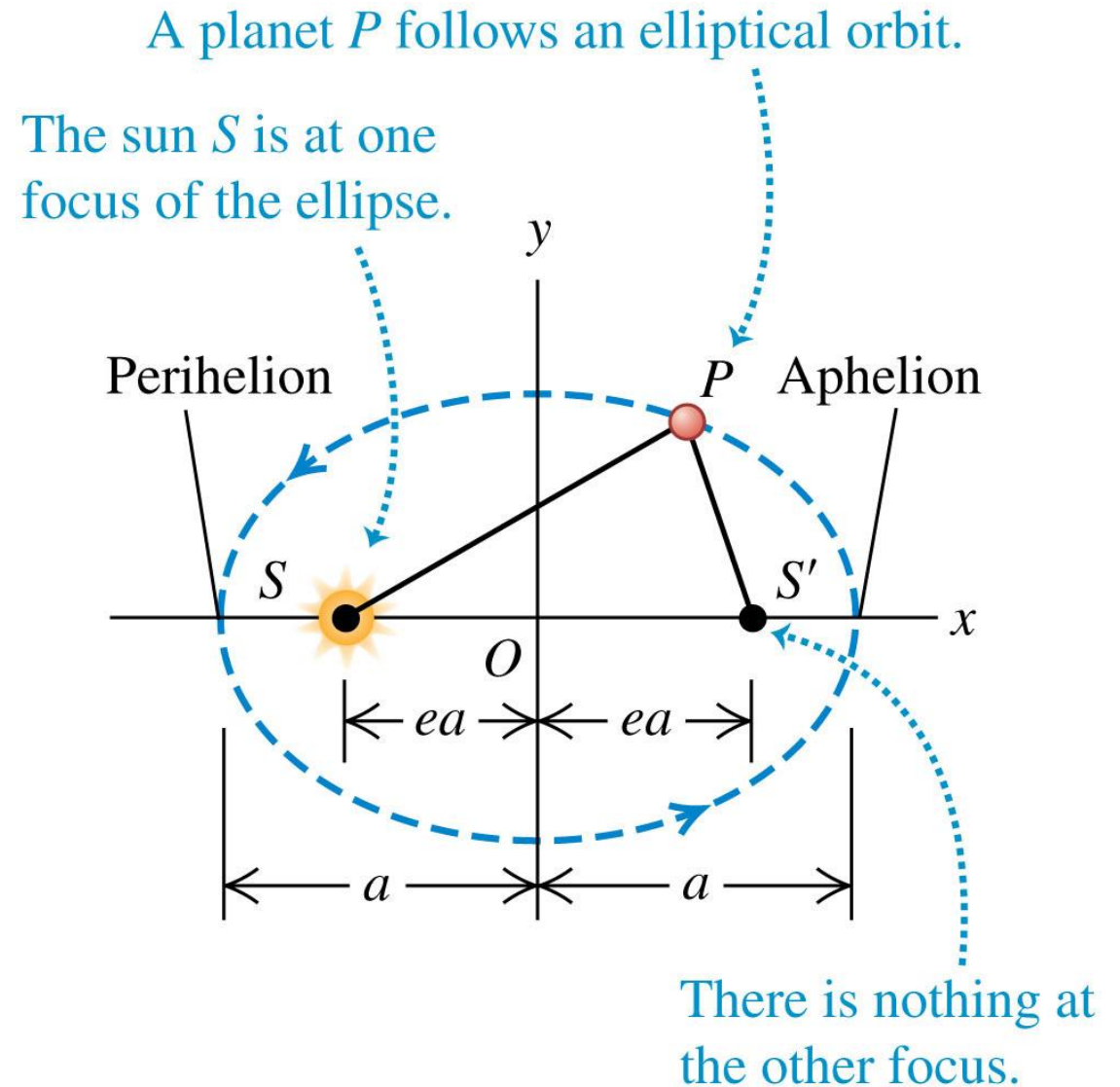


Cosmic velocities

- First cosmic velocity (orbit velocity): $v_1 = \sqrt{GM_{\oplus}/R_{\oplus}} = 7.9 \text{ km/s}$
- Second cosmic velocity (escape velocity): $v_2 = \sqrt{2GM_{\oplus}/R_{\oplus}} = 11.2 \text{ km/s}$
- Third cosmic velocity (solar system escape velocity): $v_3 = 16.7 \text{ km/s}$

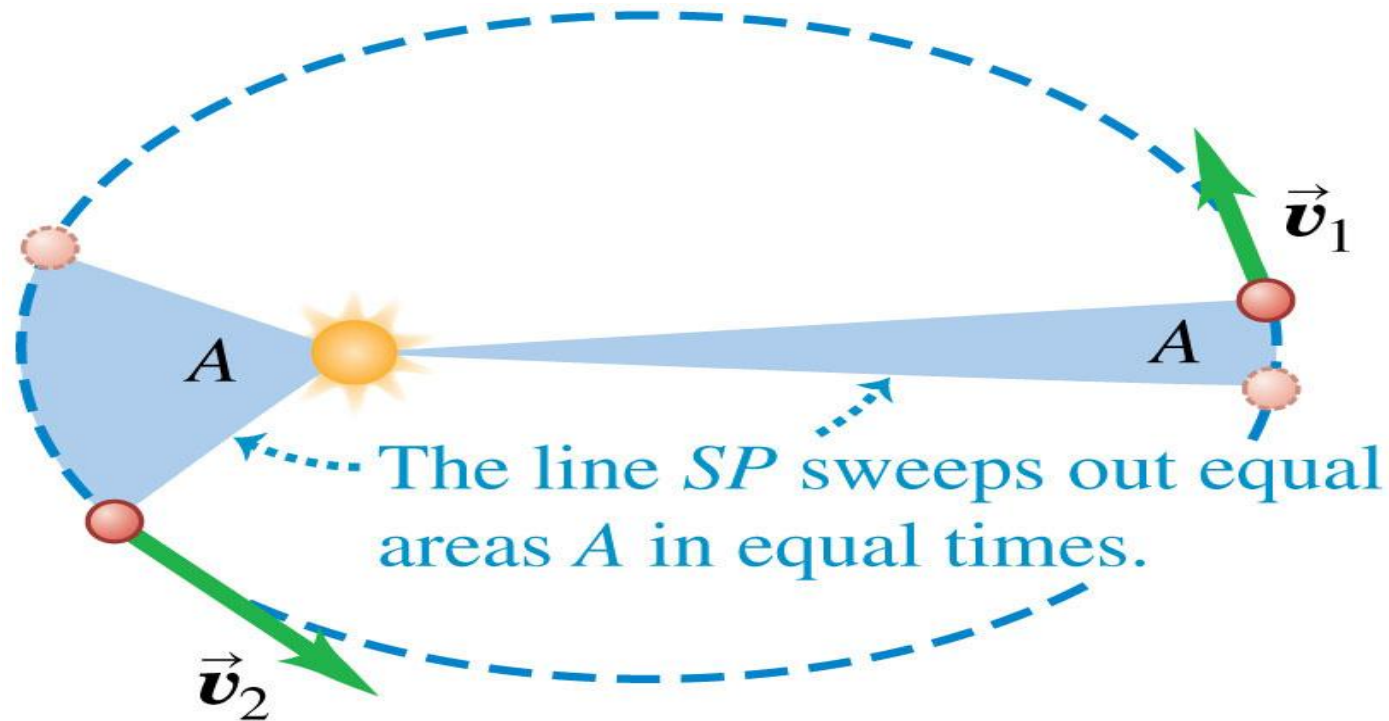
Kepler's first law

- Each planet moves in an elliptical orbit with the sun at one focus of the ellipse.



Kepler's second law

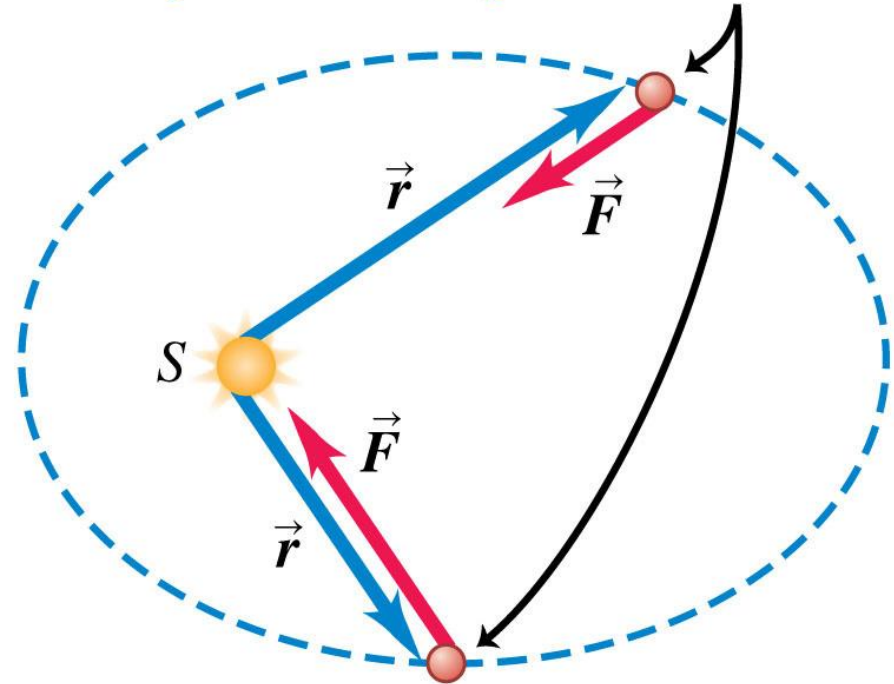
- A line from the sun to a given planet sweeps out equal areas in equal times.



Kepler's second law

- Because the gravitational force that the sun exerts on a planet produces zero torque around the sun, the planet's angular momentum around the sun remains constant.

Same planet at two points in its orbit



- Gravitational force $\vec{F}$ on planet has different magnitudes at different points but is always opposite to vector $\vec{r}$ from sun S to planet.
- Hence $\vec{F}$ produces zero torque around sun.

Kepler's third law

- The periods of the planets are proportional to the three-halves powers of the major axis lengths of their orbits.

$$T = \frac{2\pi a^{3/2}}{\sqrt{Gm_S}} \quad (\text{elliptical orbit around the sun})$$

- Note that the period does not depend on the eccentricity e .
- An asteroid in an elongated elliptical orbit with semi-major axis a will have the same orbital period as a planet in a circular orbit of radius a .

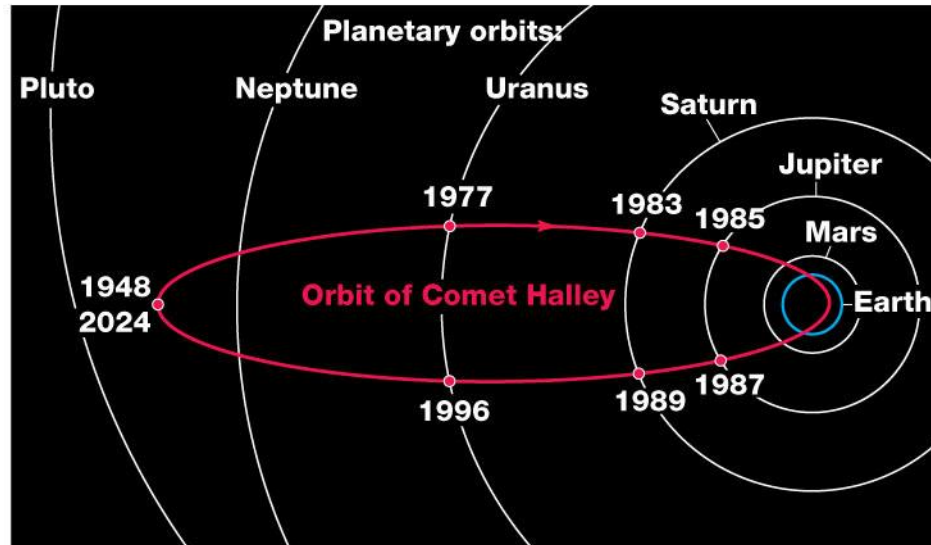
Example

Deimos is a satellite of Mars. Its distance is 23,500 km from the center of Mars, and its orbital period is 1.262 days. The mass of Mars is unknown.

- (a) Calculate the orbital velocity of Deimos.
- (b) Calculate the centripetal acceleration of Deimos.
- (c) If the radius of Mars is 3,400 km, calculate the gravitational acceleration on the surface of Mars.
- (d) Calculate the escape velocity from the surface of Mars.

Comet Halley

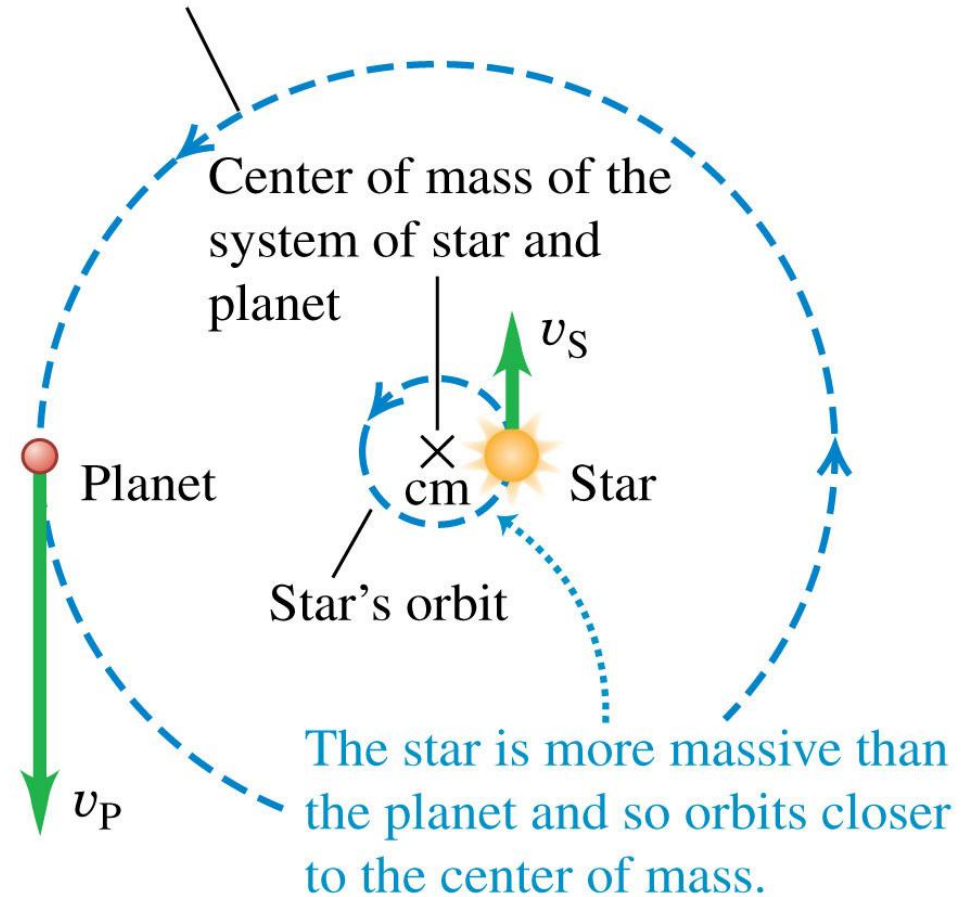
- At the heart of Comet Halley is an icy body, called the **nucleus**, that is about 10 km across.
- When the comet's orbit carries it close to the sun, the heat of sunlight causes the nucleus to partially evaporate.
- The evaporated material forms the tail, which can be tens of millions of kilometers long.



Binary system

- We have assumed that as a planet or comet orbits the sun, the sun remains absolutely stationary.
- In fact, *both* the sun and the planet orbit around their common center of mass.
- Such a system is called a binary system.

Planet's orbit around the center of mass

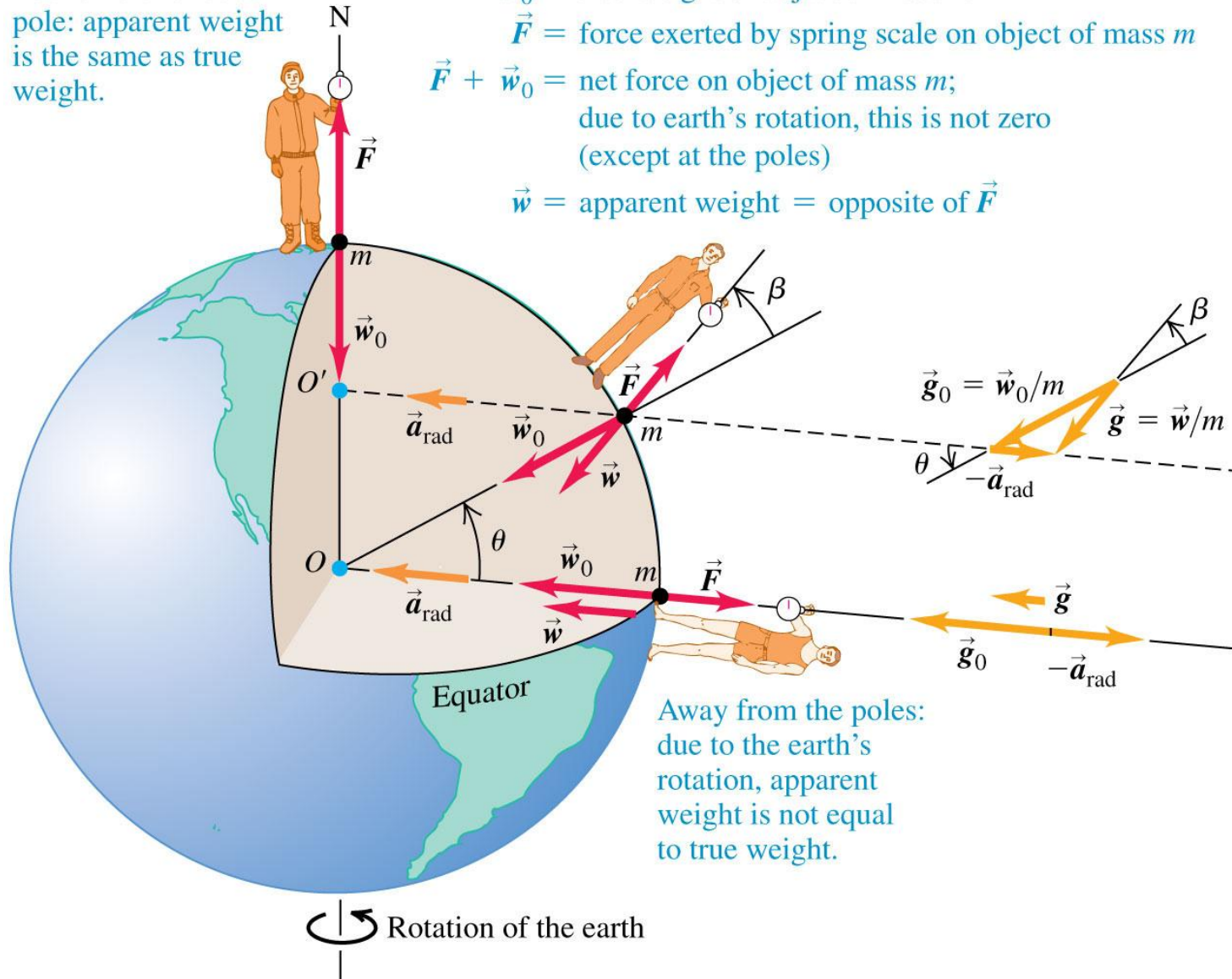


The planet and star are always on opposite sides of the center of mass.

Apparent weight and the earth's rotation

At the north or south pole: apparent weight is the same as true weight.

$\vec{w}_0$ = true weight of object of mass m
 $\vec{F}$ = force exerted by spring scale on object of mass m
 $\vec{F} + \vec{w}_0$ = net force on object of mass m ;
due to earth's rotation, this is not zero
(except at the poles)
 $\vec{w}$ = apparent weight = opposite of $\vec{F}$



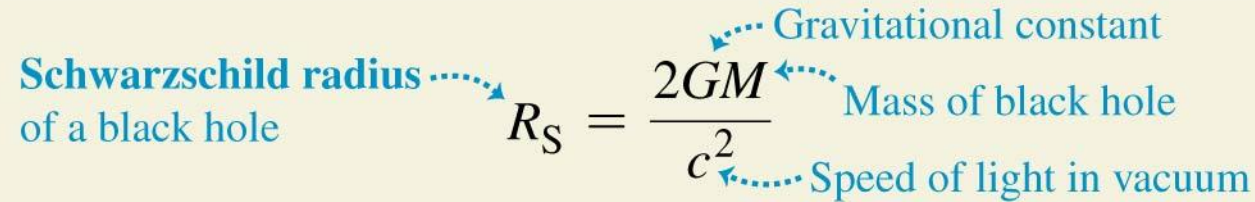
Away from the poles:
due to the earth's
rotation, apparent
weight is not equal
to true weight.

Variations of g with latitude and elevation

Station	North Latitude	Elevation (m)	g (m/s²)
Canal Zone	09°	0	9.78243
Jamaica	18°	0	9.78591
Bermuda	32°	0	9.79806
Denver, CO	40°	1638	9.79609
Pittsburgh, PA	40.5°	235	9.80118
Cambridge, MA	42°	0	9.80398
Greenland	70°	0	9.82534

Black holes

- If a spherical nonrotating body has radius less than the **Schwarzschild radius**, nothing can escape from it.
- Such a body is a **black hole**.



The diagram shows the formula for the Schwarzschild radius, $R_S = \frac{2GM}{c^2}$, with blue dotted arrows pointing from descriptive labels to the variables in the equation. The labels are: 'Schwarzschild radius of a black hole' pointing to R_S ; 'Gravitational constant' pointing to G ; 'Mass of black hole' pointing to M ; and 'Speed of light in vacuum' pointing to c^2 .

$$R_S = \frac{2GM}{c^2}$$

- The surface of the sphere with radius R_S surrounding a black hole is called the **event horizon**.
- Since light can't escape from within that sphere, we can't see events occurring inside.

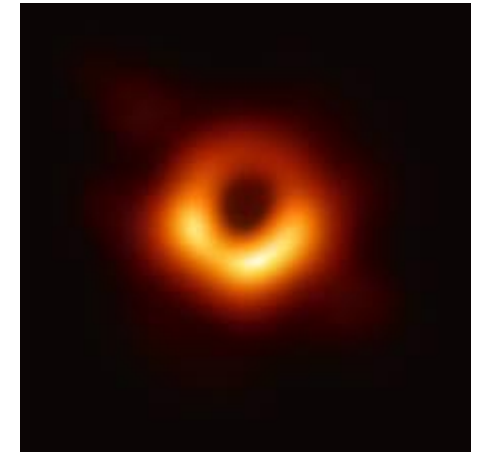
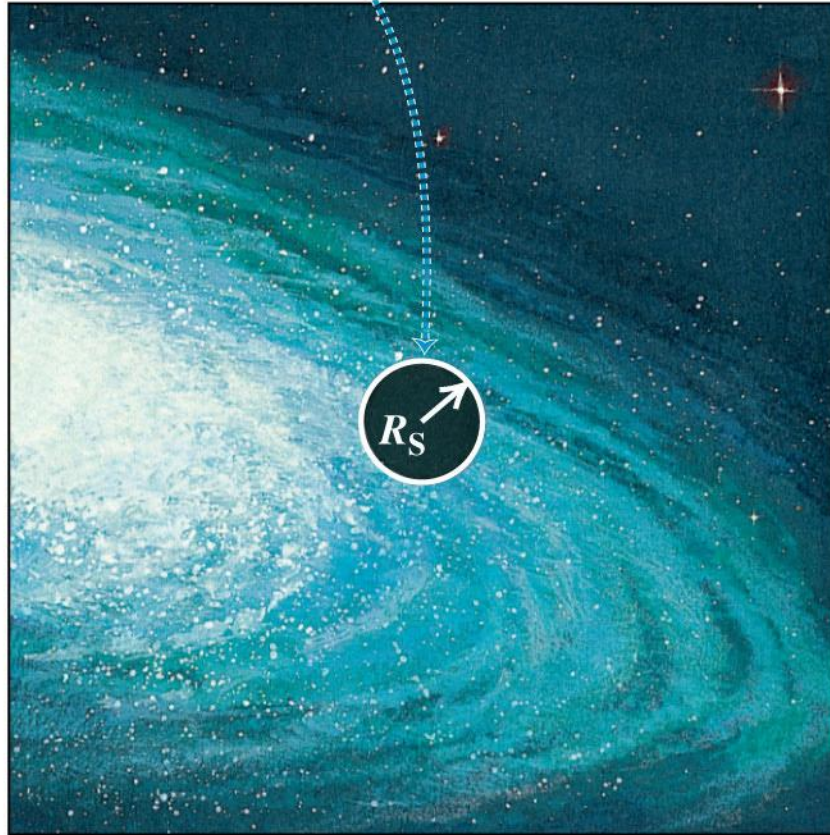
Black holes

(a) When the radius R of a body is greater than the Schwarzschild radius R_S , light can escape from the surface of the body.



Gravity acting on the escaping light “red shifts” it to longer wavelengths.

(b) If all the mass of the body lies inside radius R_S , the body is a black hole: No light can escape from it.



Event Horizon Telescope

Summary

Newton's law of gravitation

Orbit and cosmic velocities

Kepler's three laws

The change of the gravitational constant

Black holes